

Adding and subtracting vectors

Vectors can be added and subtracted in two ways. The first is by using written numbers to add the horizontal and vertical values. The second is by drawing the vectors end to end, then seeing what new vector is created.

▷ Addition

Vectors can be added in two different ways. Both methods give the same answer.

$$\begin{array}{c} \text{first vector} \\ \left(\begin{array}{c} 3 \\ 2 \end{array} \right) \end{array} + \begin{array}{c} \text{second vector} \\ \left(\begin{array}{c} -1 \\ 2 \end{array} \right) \end{array} = \begin{array}{c} 3 + (-1) = 2 \\ 2 + 2 = 4 \\ \left(\begin{array}{c} 2 \\ 4 \end{array} \right) \end{array}$$

△ Adding the parts

To add vectors numerically, add the two top numbers (the horizontal values) and then the two bottom numbers (the vertical values).

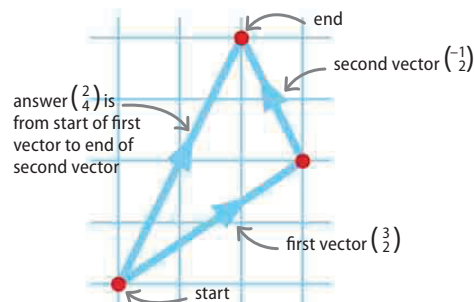
▷ Subtraction

Vectors can be subtracted in two different ways. Both methods give the same answer.

$$\begin{array}{c} \text{first vector, from} \\ \text{which second vector} \\ \text{is subtracted} \\ \left(\begin{array}{c} 3 \\ 2 \end{array} \right) \end{array} - \begin{array}{c} \text{second vector, which} \\ \text{is subtracted from} \\ \text{first vector} \\ \left(\begin{array}{c} -1 \\ 2 \end{array} \right) \end{array} = \begin{array}{c} 3 - (-1) = 4 \\ 2 - 2 = 0 \\ \left(\begin{array}{c} 4 \\ 0 \end{array} \right) \end{array}$$

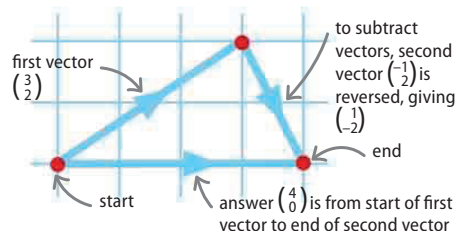
△ Subtracting the parts

To subtract one vector from another, subtract its vertical value from the vertical value of the first vector, then do the same for the horizontal values.



△ Addition by drawing vectors

Draw one vector, then draw the second starting from the end point of the first. The answer is the new vector that has been created, from the start of the first vector to the end of the second.



△ Subtraction by drawing vectors

Draw the first vector, then draw the second vector reversed, starting from the end point of the first vector. The answer to the subtraction is the vector from the start point to the end point.

Multiplying vectors

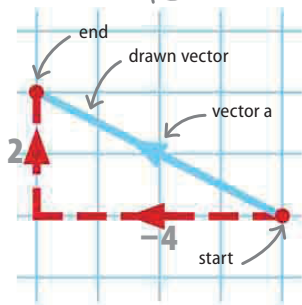
Vectors can be multiplied by numbers, but not by other vectors. The direction of a vector stays the same if it is multiplied by a positive number, but is reversed if it is multiplied by a negative number. Vectors can be multiplied by drawing or by using their numerical values.

▽ Vector a

Vector a has -4 horizontal units and +2 vertical units. It can be shown as a written vector or a drawn vector, as shown below.

$$\mathbf{a} = \left(\begin{array}{c} -4 \\ 2 \end{array} \right)$$

horizontal value
vertical value

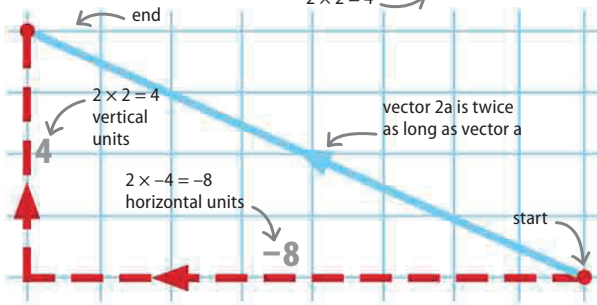


▽ Vector a multiplied by 2

To multiply vector a by 2 numerically, multiply both its horizontal and vertical parts by 2. To multiply it by 2 by drawing, simply extend the original vector by the same length again.

$$2\mathbf{a} = 2 \times \left(\begin{array}{c} -4 \\ 2 \end{array} \right) = \left(\begin{array}{c} -8 \\ 4 \end{array} \right)$$

vector a
 $2 \times -4 = -8$
 $2 \times 2 = 4$



▽ Vector a multiplied by -1/2

To multiply vector a by $-\frac{1}{2}$ numerically, multiply each of its parts by $-\frac{1}{2}$. To multiply it by $-\frac{1}{2}$ by drawing, draw a vector half the length and in the opposite direction of a.

$$-\frac{1}{2}\mathbf{a} = -\frac{1}{2} \times \left(\begin{array}{c} -4 \\ 2 \end{array} \right) = \left(\begin{array}{c} +2 \\ -1 \end{array} \right)$$

vector a
 $-\frac{1}{2} \times -4 = +2$
 $-\frac{1}{2} \times 2 = -1$

